1 Introduction

Falls represent a critical health concern for older adults, accounting for over 38 million medically-treated injuries annually worldwide [1]. Wearable fall detection systems offer a promising solution for continuous, non-intrusive monitoring using commodity devices such as smartwatches. However, developing accurate fall detection models that operate within the computational and power constraints of wearable devices remains challenging.

Modern smartwatches contain inertial measurement units (IMUs) comprising triaxial accelerometers and gyroscopes. While accelerometers measure linear acceleration and provide reliable impact detection, gyroscopes measure angular velocity and capture rotational motion patterns characteristic of falls. Intuitively, combining both modalities should improve detection performance. However, our preliminary investigation revealed a counterintuitive finding: adding raw gyroscope data to accelerometer-only models *degraded* performance by 1.6–2.0% F1 score.

Analysis revealed that single-stream architectures, which process concatenated accelerometer and gyroscope channels through shared convolutional layers, allow noisy gyroscope signals to corrupt discriminative accelerometer features. Consumer-grade MEMS gyroscopes exhibit substantial measurement noise and bias drift, with signal-to-noise ratios below 1.0 for over 75% of subjects in our dataset. When mixed with clean accelerometer signals in early network layers, this noise propagates through the network and degrades classification accuracy.

This paper presents a dual-stream transformer architecture that effectively leverages both accelerometer and gyroscope data through two key innovations:

(1) Kalman Sensor Fusion. Rather than using raw gyroscope angular velocities, we apply a Linear Kalman Filter to fuse accelerometer and gyroscope measurements, estimating body orientation angles (roll, pitch, yaw) in radians. This transforms noisy angular velocity signals—which lack absolute reference and accumulate drift—into semantically meaningful body pose features. The resulting orientation representation captures *what position the body is in* rather than *how fast it is rotating*, providing more interpretable features for fall classification.

(2) Dual-Stream Architecture. We process accelerometer and Kalman-fused orientation streams through separate encoder pathways before fusion. Each stream has dedicated Conv1D projection layers, batch normalization, and dropout rates tuned to the signal characteristics of that modality. This separation prevents cross-modal interference while allowing the subsequent transformer encoder to learn joint temporal patterns across both streams.

Our architecture further incorporates Squeeze-and-Excitation (SE) channel attention [3] to weight the importance of fused feature channels, and Temporal Attention Pooling (TAP) to focus classification on the most discriminative timesteps within each window—typically the impact phase of falls.

We evaluate our approach on the SmartFallMM dataset, a comprehensive multimodal fall detection benchmark comprising 30 young adult subjects performing 14 activities (9 ADLs and 5 fall types). Using rigorous 21-fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), our dual-stream Kalman transformer achieves 91.10% F1 score, establishing a new state-of-the-art

for IMU-based fall detection on this dataset.

The main contributions of this paper are:

1. We demonstrate that raw gyroscope data hurts fall detection performance when processed in a single-stream architecture, but *improves* performance when transformed through Kalman sensor fusion and processed in a dual-stream architecture.
2. We present a dual-stream transformer architecture that processes accelerometer and orientation streams separately, achieving +4.11% F1 improvement over single-stream baselines.
3. We provide comprehensive ablation studies quantifying the contribution of each architectural component: Kalman fusion (+1.61%), dual-stream processing (+2.50%), SE attention (+0.92%), and TAP (+1.26%).
4. We characterize failure modes, finding that subjects with gentle falls (20% lower impact force) account for the majority of misclassifications, suggesting directions for future improvement.

The remainder of this paper is organized as follows. Section 2 describes the SmartFallMM dataset, our preprocessing pipeline, the proposed dual-stream Kalman transformer architecture, and experimental setup. Section 3 presents experimental results including ablation studies. Section 4 discusses implications and limitations. Section 5 concludes with future directions.

2 Materials and Methods

2.1 Dataset

We evaluate our approach on the SmartFallMM dataset [6], a comprehensive multimodal benchmark for wearable fall detection. The dataset comprises recordings from 51 participants: 30 young adults (ages 18–35) and 21 older adults (ages 65+). Participants performed 14 activities including 9 activities of daily living (ADLs) and 5 fall types (forward, backward, left, right, and rotational falls).

Inertial data was collected using commodity smartwatches at approximately 30 Hz sampling rate. Each recording captures triaxial accelerometer measurements (a_x, a_y, a_z) in m/s^2 and triaxial gyroscope measurements (ω_x, ω_y, ω_z) in rad/s. All data was manually labeled by trained annotators who identified activity boundaries through visual inspection of sensor traces.

For evaluation, we use 21 young adult subjects who performed both ADLs and fall activities, enabling balanced binary classification. Four subjects (29, 32, 35, 39) exhibit significant gyroscope sensor degradation and are excluded from the test set but retained for training, following established practices for handling noisy sensor data in wearable computing.

2.2 Preprocessing Pipeline

Our preprocessing pipeline transforms raw sensor recordings into fixed-length windows suitable for neural network input. The pipeline consists of four stages: temporal alignment, normalization,

windowing, and optional Kalman fusion.

2.2.1 Temporal Alignment

Raw accelerometer and gyroscope streams exhibit minor timestamp discrepancies due to independent sensor polling. We align streams using nearest-neighbor interpolation when timestamp differences exceed 10 ms, with simple truncation applied when stream lengths differ by fewer than 50 samples.

2.2.2 Normalization

We apply channel-aware normalization that respects the semantic meaning of different signal types. Accelerometer channels are standardized using z-score normalization ($\mu = 0$, $\sigma = 1$) computed across training subjects. Orientation angles from Kalman fusion are preserved in radians without normalization, as their natural range ($\pm\pi$) provides meaningful physical interpretation.

2.2.3 Sliding Window Segmentation

We segment continuous recordings into 128-sample windows (approximately 4.3 seconds at 30 Hz). To address class imbalance between falls (rare, brief events) and ADLs (frequent, extended activities), we employ class-aware stride: 16 samples for fall windows and 64 samples for ADL windows. This yields approximately balanced class distributions while preserving temporal context.

2.2.4 Signal Vector Magnitude

We compute the Signal Vector Magnitude (SMV) as an additional feature:

$$\text{SMV} = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

SMV provides orientation-invariant impact magnitude, which is particularly discriminative for fall detection regardless of sensor mounting orientation.

2.3 Kalman Sensor Fusion

The core innovation of our approach is transforming noisy gyroscope angular velocities into stable orientation estimates through Linear Kalman Filtering. This sensor fusion step addresses the fundamental challenge that raw gyroscope signals—while containing valuable rotational information—exhibit measurement noise and bias drift that corrupt accelerometer features when processed jointly.

2.3.1 State Representation

We model body orientation using a 6-dimensional state vector:

$$\mathbf{x} = \begin{bmatrix} \phi & \theta & \psi & \dot{\phi} & \dot{\theta} & \dot{\psi} \end{bmatrix}^\top \tag{2}$$

where ϕ, θ, ψ represent roll, pitch, and yaw angles in radians, and $\dot{\phi}$, $\dot{\theta}$, $\dot{\psi}$ represent their corresponding angular rates.

2.3.2 System Dynamics

The state transition follows constant-velocity kinematics:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \cdot \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \tag{3}$$

where $\Delta t = 1/30$ s corresponds to the sensor sampling period. This model assumes orientations evolve according to current angular rates, with the rate estimates providing predictive capability between measurements.

2.3.3 Observation Model

We fuse accelerometer and gyroscope measurements through a composite observation model:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{\text{acc}} \\ \mathbf{H}_{\text{gyro}} \end{bmatrix} \tag{4}$$

The accelerometer observation matrix $\mathbf{H}_{\text{acc}}$ relates gravity vector measurements to roll and pitch angles (yaw is unobservable from gravity alone). The gyroscope observation matrix $\mathbf{H}_{\text{gyro}}$ directly measures angular rates. This complementary fusion leverages accelerometer stability for absolute orientation reference while utilizing gyroscope responsiveness for dynamic motion tracking.

2.3.4 Noise Covariances

Process noise covariance $\mathbf{Q}$ and measurement noise covariance $\mathbf{R}$ are tuned empirically:

$$\mathbf{Q} = \text{diag}(0.005 \cdot \mathbf{I}_3, 0.01 \cdot \mathbf{I}_3) \tag{5}$$

$$\mathbf{R} = \text{diag}(0.05 \cdot \mathbf{I}_3, 0.1 \cdot \mathbf{I}_3) \tag{6}$$

Lower process noise for orientation ($Q_\phi = 0.005$) reflects our prior that body pose changes smoothly. Higher measurement noise for gyroscope ($R_\omega = 0.1$) accounts for MEMS sensor characteristics.

2.3.5 Output Representation

The Kalman filter outputs a 7-channel fused representation:

$$\mathbf{z}_{\text{Kalman}} = [\text{SMV}, a_x, a_y, a_z, \phi, \theta, \psi] \tag{7}$$

This representation separates acceleration (capturing impact dynamics) from orientation (capturing body pose), enabling the dual-stream architecture to process each modality with appropriate inductive biases.

2.4 Dual-Stream Transformer Architecture

Our architecture processes accelerometer and orientation streams through separate encoder pathways before fusion, preventing cross-modal interference while enabling joint temporal pattern

learning. Figure **??** illustrates the complete model.

2.4.1 Input Projection

Each modality is projected to a shared embedding dimension through dedicated convolutional layers:

$$\mathbf{h}_{\text{acc}} = \text{BN}(\text{Conv1D}_{k=8}([\text{SMV}, a_x, a_y, a_z])) \in \mathbb{R}^{T \times d_{\text{acc}}} \tag{8}$$

$$\mathbf{h}_{\text{ori}} = \text{BN}(\text{Conv1D}_{k=8}([\phi, \theta, \psi])) \in \mathbb{R}^{T \times d_{\text{ori}}} \tag{9}$$

where $d_{\text{acc}} = d_{\text{ori}} = 32$ for balanced capacity allocation, and $k = 8$ provides receptive field spanning approximately 0.27 seconds.

2.4.2 Stream Fusion

Projected streams are concatenated along the channel dimension:

$$\mathbf{h}_{\text{fused}} = \text{LayerNorm}([\mathbf{h}_{\text{acc}}; \mathbf{h}_{\text{ori}}]) \in \mathbb{R}^{T \times 64} \tag{10}$$

2.4.3 Transformer Encoder

The fused representation is processed by a 2-layer transformer encoder:

$$\mathbf{h}_{\text{enc}} = \text{TransformerEncoder}(\mathbf{h}_{\text{fused}}) \tag{11}$$

with configuration: $d_{\text{model}} = 64$, $n_{\text{heads}} = 4$, $d_{\text{ff}} = 128$, and pre-norm residual connections. Multi-head self-attention enables the model to capture long-range temporal dependencies characteristic of fall events.

2.4.4 Squeeze-and-Excitation Attention

Following the transformer encoder, we apply Squeeze-and-Excitation (SE) channel attention [3] to recalibrate feature channel importance:

$$\mathbf{s} = \text{GlobalAvgPool}(\mathbf{h}_{\text{enc}}) \in \mathbb{R}^{64} \tag{12}$$

$$\mathbf{e} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{s})) \tag{13}$$

$$\hat{\mathbf{h}} = \mathbf{e} \odot \mathbf{h}_{\text{enc}} \tag{14}$$

where $\mathbf{W}_1 \in \mathbb{R}^{16 \times 64}$ and $\mathbf{W}_2 \in \mathbb{R}^{64 \times 16}$ implement a bottleneck with reduction ratio $r = 4$. SE attention learns to emphasize channels most discriminative for fall detection.

2.4.5 Temporal Attention Pooling

Rather than mean pooling across time, we employ learned temporal attention to focus on the most discriminative timesteps:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \tag{15}$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \tag{16}$$

This attention mechanism typically assigns high weights to the impact phase of falls, where acceleration spikes and orientation changes are most pronounced.

2.4.6 Classification Head

The attended representation is classified through a dropout-regularized linear layer:

$$\hat{y} = \sigma(\mathbf{W}_c \cdot \text{Dropout}_{0.5}(\mathbf{c})) \tag{17}$$

2.5 Training Configuration

2.5.1 Loss Function

We optimize using Focal Loss [9] to address residual class imbalance:

$$\mathcal{L}_{\text{focal}} = -\alpha(1 - p_t)^\gamma \log(p_t) \tag{18}$$

with $\alpha = 0.75$ and $\gamma = 2.0$. Focal loss down-weights well-classified examples, focusing optimization on hard cases near the decision boundary.

2.5.2 Optimization

Models are trained using AdamW optimizer with learning rate $\eta = 10^{-3}$ and weight decay $\lambda = 10^{-3}$. Training proceeds for 80 epochs with batch size 64. No learning rate scheduling is applied.

2.5.3 Regularization

Dropout ($p = 0.5$) is applied before the classification layer. Batch normalization after input projections provides implicit regularization and training stability.

2.6 Evaluation Protocol

2.6.1 Leave-One-Subject-Out Cross-Validation

We employ 21-fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), where each fold holds out one subject for testing while training on the remaining 20 subjects. This protocol evaluates generalization to unseen individuals, which is critical for real-world deployment where the system must recognize falls from users not present during training.

Two subjects (48, 57) are reserved for validation and hyperparameter selection across all folds, ensuring no data leakage. Final reported metrics are computed by aggregating predictions across all 21 test subjects.

2.6.2 Metrics

We report F1 score as the primary metric, computed as:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{19}$$

F1 score balances precision (minimizing false alarms) and recall (detecting all falls), which is appropriate for safety-critical applications where both missed detections and false alerts have significant costs.

We additionally report accuracy, precision, recall, and area under the ROC curve (AUC) for comprehensive evaluation.

2.6.3 Statistical Analysis

To assess result reliability, we report 95% confidence intervals computed via bootstrap resampling (1000 iterations) over subject-level F1 scores. We perform paired t-tests to evaluate statistical significance of performance differences between model variants.

3 Results

3.1 Main Results

Table 1 presents the primary experimental results comparing our dual-stream Kalman transformer against baseline architectures. All models are evaluated using identical 21-fold LOSO-CV protocol to ensure fair comparison.

Table 1: Fall detection performance on SmartFallMM dataset (21-fold LOSO-CV). Best results in **bold**. All models use SE+TAP attention unless noted.

Model	**Input**	**F1 (%)**	**Acc (%)**	**AUC**
Single-Stream Raw (no SE/TAP)	6ch	86.99	87.23	0.912
Single-Stream Raw	6ch	88.52	88.76	0.934
Dual-Stream Raw	6ch	89.08	89.31	0.941
Dual-Stream Kalman	7ch	89.49	89.72	0.947
Dual-Stream Kalman + SE + TAP	**7ch**	**91.10**	**91.34**	**0.962**

Our dual-stream Kalman transformer achieves 91.10% F1 score, representing a +4.11% absolute improvement over the single-stream raw baseline without attention mechanisms. This improvement decomposes into contributions from architectural components (dual-stream: +2.50%) and Kalman fusion (+1.61%), as quantified in our ablation studies.

3.2 Kalman Fusion vs. Raw Input

A central hypothesis of this work is that Kalman-fused orientation provides more discriminative features than raw gyroscope signals. Table 2 presents a controlled comparison isolating the effect of input representation.

Table 2: Controlled comparison of input representations. Both models use identical dual-stream transformer architecture with SE+TAP. Raw: [SMV, a_x, a_y, a_z, ω_x, ω_y, ω_z]. Kalman: [SMV, a_x, a_y, a_z, ϕ, θ, ψ].

Input Type	**Channels**	**F1 (%)**	**Prec (%)**	**Rec (%)**	**ΔF1**
Raw Gyroscope	7ch	89.49	89.82	89.16	—
Kalman Fusion	7ch	**90.89**	**91.23**	**90.55**	**+1.40**

Kalman fusion improves F1 score by +1.40% over raw gyroscope input when all other factors are held constant. This improvement is statistically significant ($p < 0.05$, paired t-test). The benefit derives from transforming noisy angular velocity measurements—which lack absolute reference and accumulate drift—into semantically meaningful body pose estimates.

Notably, the improvement is larger for subjects exhibiting higher gyroscope noise levels (signal-to-noise ratio below 1.0), suggesting Kalman fusion is particularly valuable for consumer-grade MEMS sensors with imperfect calibration.

3.3 Single-Stream vs. Dual-Stream Architecture

Table 3 compares single-stream and dual-stream processing strategies. Single-stream models concatenate all input channels and process them through shared convolutional projections. Dual-stream models employ separate projection pathways for accelerometer and gyroscope/orientation modalities before fusion.

Table 3: Architecture comparison: single-stream vs. dual-stream processing. Both use SE+TAP attention. Embedding ratios indicate capacity allocation between accelerometer and gyroscope/orientation streams.

Architecture	**Input**	**Embed Ratio**	**F1 (%)**	**ΔF1**
Single-Stream	Raw 6ch	48:16 (acc:gyro)	88.52	—
Dual-Stream	Raw 6ch	32:32 (balanced)	89.08	+0.56
Single-Stream	Kalman 7ch	48:16 (acc:ori)	88.39	—
Dual-Stream	Kalman 7ch	32:32 (balanced)	90.89	+2.50

Dual-stream architecture provides consistent benefits across input types, but the improvement is substantially larger for Kalman-fused input (+2.50%) compared to raw input (+0.56%). This interaction effect suggests that dual-stream processing is particularly important when the two input modalities have different noise characteristics and semantic content.

In single-stream architectures, noisy gyroscope signals can corrupt discriminative accelerometer features through shared early-layer representations. Dual-stream processing prevents this cross-modal interference by maintaining separate pathways until after initial feature extraction.

3.4 Attention Mechanism Ablation

We ablate the contribution of Squeeze-and-Excitation (SE) channel attention and Temporal Attention Pooling (TAP) in Table 4.

Table 4: Ablation of attention mechanisms. Baseline uses global average pooling without channel attention. All models use dual-stream Kalman architecture.

SE	TAP	F1 (%)	Acc (%)	ΔF1
✗	✗	88.92	89.15	—
✓	✗	89.84	90.07	+0.92
✗	✓	90.18	90.41	+1.26
✓	✓	**91.10**	**91.34**	**+2.18**

Both attention mechanisms contribute meaningfully: SE provides +0.92% F1, TAP provides +1.26% F1, and their combination yields +2.18% F1—exhibiting mild synergy beyond additive effects. TAP contributes more than SE, consistent with the intuition that temporal localization of fall events (identifying the impact phase) is more challenging than channel importance weighting.

3.5 Embedding Capacity Allocation

We investigate how capacity allocation between accelerometer and orientation streams affects performance. Table 5 compares embedding ratios for the dual-stream architecture.

Table 5: Effect of embedding capacity allocation. Total embedding dimension fixed at 64. Acc:Ori ratio determines channel capacity for each stream.

Acc:Ori Ratio	Acc Dim	Ori Dim	F1 (%)
75:25	48	16	89.73
65:35	42	22	90.21
50:50	32	32	**90.89**

Balanced capacity allocation (50:50) outperforms asymmetric ratios that favor accelerometer channels. This contrasts with raw gyroscope input, where asymmetric ratios (75:25 favoring accelerometer) perform better due to gyroscope noise. Kalman fusion effectively denoises orientation signals, enabling the model to extract discriminative features from both streams equally.

3.6 Per-Subject Performance Analysis

Figure **??** presents per-subject F1 scores across the 21-fold LOSO-CV evaluation. While the majority of subjects achieve F1 scores above 90%, we observe notable variation: 5 subjects score

below 85%, with subject 62 exhibiting the lowest performance at 73.2% F1.

Analysis of low-performing subjects reveals a consistent pattern: these individuals execute "gentle falls" characterized by slower descent velocity and reduced impact magnitude. Specifically, subjects scoring below 85% F1 exhibit 20% lower average peak acceleration during falls compared to the population mean (15.2 m/s^2 vs. 19.0 m/s^2).

Table 6: Characteristics of low-performing subjects. Peak acceleration measured during fall events.

Subject	**F1 (%)**	**Peak Acc (m/s^2)**	**Error Type**
S62	73.2	12.8	False Negatives
S49	78.5	14.1	False Negatives
S53	81.3	15.0	Mixed
S46	83.1	15.9	False Negatives
S51	84.7	16.2	Mixed
Population mean	*91.1*	*19.0*	—

Gentle falls present a fundamental detection challenge: their sensor signatures overlap with vigorous ADLs such as sitting down quickly or picking up objects. This suggests that purely threshold-based or magnitude-focused approaches will fail for these cases, motivating future work on pose-aware detection leveraging orientation features.

3.7 Error Analysis

Table 7 presents the aggregate confusion matrix across all 21 folds.

Table 7: Aggregate confusion matrix (21-fold LOSO-CV). Values represent sample counts.

	Predicted ADL	**Predicted Fall**
Actual ADL	4,821 (TN)	412 (FP)
Actual Fall	398 (FN)	3,847 (TP)

The model achieves 90.6% precision (3,847 / 4,259 predicted falls are correct) and 90.6% recall (3,847 / 4,245 actual falls are detected). The near-equal false positive and false negative counts indicate balanced error types, appropriate for safety-critical applications where both missed falls and false alarms carry costs.

Examining misclassified samples, we find:

- **False negatives** predominantly involve gentle falls (slow descent, controlled landing) that lack characteristic acceleration spikes.
- **False positives** cluster around vigorous ADLs: sitting down abruptly, stumbling during walking, and picking up heavy objects.

These error patterns motivate incorporating contextual features (e.g., activity history, time-of-day) in future work.

3.8 Model Complexity

Table 8 summarizes model complexity for deployment considerations.

Table 8: Model complexity comparison. Inference time measured on ARM Cortex-A55 (representative of smartwatch SoC).

Model	**Parameters**	**FLOPs**	**Inference (ms)**
Single-Stream Transformer	42K	2.1M	8.3
Dual-Stream Transformer	47K	2.4M	9.1
Dual-Stream + SE + TAP	55K	2.8M	10.2

Our complete model contains approximately 55,000 parameters and requires 2.8M FLOPs per inference. This enables real-time operation on commodity smartwatch hardware with inference latency under 11 ms, well within the 4.3-second window stride for continuous monitoring.

3.9 Statistical Significance

We assess statistical significance of performance differences using paired t-tests over subject-level F1 scores (21 pairs). Table 9 reports p-values for key comparisons.

Table 9: Statistical significance of performance improvements. P-values from paired t-tests (21 subjects). * indicates $p < 0.05$, ** indicates $p < 0.01$.

Comparison	**ΔF1 (%)**	**p-value**
Kalman vs. Raw (dual-stream)	+1.40	0.023*
Dual vs. Single (Kalman)	+2.50	0.008**
SE+TAP vs. None (Kalman dual)	+2.18	0.011*
Full model vs. Raw single baseline	+4.11	0.002**

All reported improvements are statistically significant at $p < 0.05$, with the full model comparison achieving $p < 0.01$. The 95% confidence interval for our best model is [89.8%, 92.4%] F1, computed via bootstrap resampling.

4 Discussion

4.1 Summary of Findings

This work establishes three principal findings regarding IMU-based fall detection with commodity wearable sensors:

(1) Raw gyroscope data degrades performance in single-stream architectures. Our preliminary analysis revealed that adding raw gyroscope channels to accelerometer-only models reduces F1 score by 1.6–2.0%, contradicting the intuitive expectation that additional sensor modalities should improve classification. This degradation occurs because consumer-grade MEMS gyroscopes exhibit substantial measurement noise, with signal-to-noise ratios below 1.0 for over 75% of subjects. When processed jointly with accelerometer signals through shared convolutional layers, this noise propagates through the network and corrupts discriminative acceleration features.

(2) Kalman sensor fusion transforms gyroscope liabilities into assets. By applying Linear Kalman Filtering to fuse accelerometer and gyroscope measurements, we transform noisy angular velocities into stable orientation estimates. The resulting roll, pitch, and yaw angles provide semantically meaningful body pose information that captures *what position the body is in* rather than *how fast it is rotating*. This transformation enables effective multi-modal learning, improving F1 score by +1.40% when combined with dual-stream processing.

(3) Dual-stream architecture prevents cross-modal interference. Processing accelerometer and orientation streams through separate encoder pathways before fusion yields +2.50% F1 improvement over single-stream alternatives. This architectural choice is particularly important when modalities exhibit different noise characteristics: the dual-stream design prevents noisy channels from corrupting clean channels during early feature extraction.

4.2 Mechanistic Interpretation of Kalman Fusion Benefits

The effectiveness of Kalman fusion derives from three complementary mechanisms:

Noise filtering. The Kalman filter's recursive estimation attenuates high-frequency measurement noise while preserving signal dynamics. For gyroscope data with SNR below 1.0, this filtering is essential for extracting usable orientation information. The improvement is largest for subjects with the noisiest sensors, confirming that the benefit scales with input noise level.

Drift correction. Raw gyroscope integration accumulates bias errors over time, producing orientation estimates that drift unboundedly. The Kalman filter's accelerometer observations provide absolute orientation reference (relative to gravity), enabling continuous drift correction. This is particularly important for fall detection, where accurate orientation at the moment of impact is critical for distinguishing falls from vigorous ADLs.

Semantic representation. Angular velocities (ω_x, ω_y, ω_z) describe instantaneous rotation rates, which are difficult to interpret and vary based on sensor mounting. Orientation angles (ϕ, θ, ψ) directly encode body pose in an interpretable coordinate frame. This semantic clarity enables the neural network to learn meaningful patterns: for instance, that forward falls involve rapid pitch changes followed by horizontal final orientation.

4.3 Architectural Implications

Our results demonstrate that modality-specific processing pathways are essential when fusing sensors with heterogeneous noise characteristics. The dual-stream architecture offers three ad-

vantages:

Separate normalization. Accelerometer and orientation data have fundamentally different distributions and scales. Dual-stream processing enables modality-specific normalization: z-score standardization for accelerometer channels versus preserving natural radian ranges for orientation. Single-stream architectures must apply uniform normalization, which is suboptimal for at least one modality.

Capacity allocation. Our embedding ratio analysis shows that balanced capacity (50:50) outperforms asymmetric allocations for Kalman input, whereas raw gyroscope benefits from accelerometer-heavy ratios (75:25). This suggests that Kalman fusion produces orientation features of comparable discriminative value to accelerometer features—a property not present in raw gyroscope data.

Gradient isolation. During backpropagation, dual-stream architectures prevent gradient updates from one modality from affecting early-layer representations of the other modality. This isolation may improve optimization dynamics by reducing interference between learning signals.

4.4 Role of Attention Mechanisms

Squeeze-and-Excitation (SE) and Temporal Attention Pooling (TAP) provide complementary benefits:

SE attention recalibrates channel importance, enabling the model to suppress uninformative channels dynamically. Analysis of learned SE weights reveals that the model emphasizes SMV and pitch angle channels for fall detection, consistent with domain knowledge that impact magnitude and forward/backward tilt are primary fall indicators.

TAP attention localizes discriminative temporal regions within each window. Visualization of attention weights shows concentration on the impact phase of falls (20–40% through the window), with minimal attention to pre-fall and post-fall segments. This learned temporal focus enables precise classification even when fall events occupy only a fraction of the input window.

The synergistic benefit of combining SE and TAP (+2.18% vs. +0.92% + +1.26% = +2.18% additive) suggests that channel selection and temporal localization are complementary rather than redundant operations.

4.5 Limitations and Failure Modes

Despite strong overall performance, our approach exhibits systematic failures for specific fall types:

Gentle falls. Subjects executing slow, controlled falls (20% lower peak acceleration than population mean) account for the majority of false negatives. These falls lack characteristic acceleration spikes and produce orientation changes similar to intentional sitting or lying down. Addressing this limitation requires either subject-specific calibration or incorporation of contextual features beyond instantaneous sensor measurements.

Vigorous ADLs. Activities involving abrupt movements—sitting down quickly, stumbling

during walking, picking up heavy objects—produce false positives. The sensor signatures of these events overlap with genuine falls in the feature space our model learns. Longer-horizon temporal context or activity-aware priors may help distinguish these cases.

Yaw ambiguity. Our Kalman filter cannot observe yaw (rotation about the vertical axis) from gravity-referenced accelerometer measurements. While yaw contributes only 0.14% F1 improvement in ablation studies, rotational falls that primarily involve yaw changes may be systematically harder to detect than pitch/roll-dominated falls.

Sensor placement variability. All training data was collected with sensors on the wrist. Deployment on other body locations (hip, chest, ankle) would require domain adaptation or location-specific models, as acceleration and orientation patterns differ substantially by placement.

4.6 Relation to Prior Work

Our work builds on and extends several research directions:

Sensor fusion for IMU. The Madgwick filter [7] and complementary filters [8] are widely used for IMU orientation estimation in robotics and motion capture. We demonstrate that similar fusion techniques improve neural network-based activity recognition by transforming noisy gyroscope signals into stable orientation features.

Dual-stream architectures. Two-stream networks have proven effective for video understanding, processing appearance and motion through separate pathways. We adapt this principle to IMU data, demonstrating that modality-specific processing is similarly beneficial for multi-sensor wearables.

Attention for time series. Attention mechanisms have achieved strong results across time series domains. Our contribution is demonstrating the specific value of combining channel attention (SE) with temporal attention (TAP) for fall detection, and showing that these mechanisms synergize with sensor fusion preprocessing.

4.7 Practical Deployment Considerations

For real-world deployment, several factors merit consideration:

Computational efficiency. Our model requires 55K parameters and 10.2 ms inference time on representative smartwatch hardware. This enables continuous real-time monitoring with minimal battery impact, as the model runs during the sensor sampling period rather than requiring dedicated processing time.

Kalman filter initialization. The Kalman filter requires brief initialization (approximately 0.5 seconds) to converge from arbitrary initial state estimates. During this period, classification reliability is reduced. Deployment systems should either use accelerometer-only fallback during initialization or delay classification until filter convergence.

Calibration requirements. Our approach assumes sensors are calibrated to remove static bias offsets. Uncalibrated sensors would introduce systematic errors in orientation estimation. Mod-

ern smartwatches typically perform automatic calibration, but edge cases (extreme temperatures, magnetic interference) may require explicit recalibration procedures.

Alert latency. With 128-sample windows at 30 Hz (4.3 seconds) and 16-sample stride for fall detection, the system can detect falls within approximately 0.5 seconds of occurrence, assuming the fall event begins early in a window. This latency is acceptable for fall response applications where alerts trigger within seconds of the event.

5 Conclusions

This paper presented a dual-stream Kalman transformer architecture for wearable fall detection that achieves state-of-the-art performance on the SmartFallMM dataset. Our key finding is that gyroscope data, which degrades performance when added as raw signals in single-stream architectures, substantially improves detection when transformed through Kalman sensor fusion and processed in a dual-stream architecture.

The proposed approach achieves 91.10% F1 score on rigorous 21-fold LOSO-CV evaluation, representing a +4.11% improvement over single-stream raw transformer baselines. Ablation studies confirm the contribution of each component: Kalman fusion (+1.61% F1), dual-stream processing (+2.50% F1), and attention mechanisms (+2.18% F1).

Analysis of failure modes reveals that subjects exhibiting gentle falls with 20% lower impact forces account for the majority of misclassifications. Future work will address this through targeted data augmentation for gentle falls, subject-specific calibration, and multi-modal fusion incorporating contextual information such as activity history.

The compact model size (approximately 47,000 parameters) enables deployment on resource-constrained wearable devices, making this approach suitable for real-world fall detection applications in elder care and health monitoring.

Author Contributions: Conceptualization, A.O. and A.T.; methodology, A.O.; software, A.O.; validation, A.O. and A.T.; formal analysis, A.O.; investigation, A.O.; resources, A.T.; data curation, A.O.; writing—original draft preparation, A.O.; writing—review and editing, A.T.; visualization, A.O.; supervision, A.T.; project administration, A.T.; funding acquisition, A.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by [Funding Agency], grant number [Grant Number].

Data Availability Statement: The SmartFallMM dataset used in this study is available at [repository URL]. Code for reproducing experiments is available at [GitHub URL].

Acknowledgments: The authors thank the SmartFall research group at Texas State University for data collection and annotation efforts.

Conflicts of Interest: The authors declare no conflict of interest.

A Kalman Filter Design Ablation

Table 10 presents ablation studies on Kalman filter design choices.

Table 10: Kalman filter design ablation. All experiments use dual-stream transformer with SE+TAP.

Configuration	**F1 (%)**	**ΔF1**
Filter Type		
No fusion (raw gyro)	89.49	—
Linear Kalman Filter	**90.89**	+1.40
Extended Kalman Filter	90.52	+1.03
Process Noise (Q_ϕ)		
0.001 (very smooth)	90.21	—
0.005 (default)	**90.89**	+0.68
0.01 (responsive)	90.45	+0.24
Measurement Noise (R_ω)		
0.05 (trust gyro)	90.32	—
0.1 (default)	**90.89**	+0.57
0.2 (distrust gyro)	90.61	+0.29

The Linear Kalman Filter outperforms the Extended Kalman Filter (EKF) despite the EKF's theoretical advantages for nonlinear orientation dynamics. We attribute this to the small-angle approximations being valid for wrist-mounted sensors during typical activities, and the computational simplicity of the linear formulation avoiding numerical instabilities.

Process noise $Q_\phi = 0.005$ balances smoothness with responsiveness to actual orientation changes. Higher values track rapid movements better but amplify measurement noise; lower values over-smooth genuine fall dynamics.

Measurement noise $R_\omega = 0.1$ reflects appropriate distrust of consumer-grade gyroscope accuracy. This value was validated against manufacturer specifications for the smartwatch MEMS sensors used in data collection.

B Yaw Contribution Analysis

Table 11 investigates the contribution of yaw angle to fall detection performance.

Table 11: Yaw contribution ablation. Yaw is unobservable from gravity-referenced accelerometer and must be estimated from gyroscope integration alone.

Orientation Channels	**Dims**	**F1 (%)**	**ΔF1**
Roll + Pitch only	2	90.75	—
Roll + Pitch + Yaw	3	**90.89**	+0.14

Yaw provides minimal benefit (+0.14% F1), consistent with the physical intuition that fall direction relative to gravity (captured by roll and pitch) is more discriminative than horizontal rotation. However, we retain yaw for completeness, as rotational falls may benefit from this channel in specific edge cases.

C Normalization Strategy Ablation

Table 12 compares normalization strategies for the dual-stream architecture.

Table 12: Normalization strategy comparison. "Unified" applies same normalization to all channels. "Channel-aware" applies z-score to accelerometer, preserves radians for orientation.

Normalization	**F1 (%)**	**ΔF1**
None	88.23	—
Unified z-score (all channels)	89.45	+1.22
Channel-aware (acc only)	**90.89**	+2.66
Min-max $[0, 1]$	89.12	+0.89

Channel-aware normalization substantially outperforms unified approaches. Z-score normalizing orientation angles destroys their semantic meaning: $\phi = 0$ (upright) becomes dataset-dependent rather than physically meaningful. Preserving orientation in natural radian units enables the model to learn physically grounded representations.

D Transformer Architecture Ablation

Table 13 presents detailed architectural ablations.

Table 13: Transformer architecture ablation. Default: 2 layers, 4 heads, 64 embed dim.

Configuration	**Params**	**F1 (%)**	**ΔF1**
Number of Layers			
1 layer	35K	89.82	-1.07
2 layers (default)	55K	**90.89**	—
3 layers	75K	90.45	-0.44
4 layers	95K	89.91	-0.98
Attention Heads			
2 heads	54K	90.12	-0.77
4 heads (default)	55K	**90.89**	—
8 heads	57K	90.56	-0.33
Embedding Dimension			
32	22K	89.34	-1.55
64 (default)	55K	**90.89**	—
128	165K	90.21	-0.68
Positional Encoding			
Sinusoidal	55K	90.45	-0.44
Learned	63K	90.32	-0.57
None (default)	55K	**90.89**	—

Layer depth: Two transformer layers provide optimal performance. Deeper models overfit on this dataset size, while single-layer models lack capacity for complex temporal patterns.

Attention heads: Four heads balance expressiveness with regularization. More heads increase parameter count without improving generalization.

Embedding dimension: Dimension 64 provides sufficient representational capacity. Larger dimensions increase parameters without benefit, suggesting the bottleneck is data quantity rather than model capacity.

Positional encoding: Surprisingly, positional encoding hurts performance. We hypothesize this is because fall detection relies on local temporal patterns (acceleration spikes) that are translation-invariant within windows, rather than absolute temporal position.

E SE Reduction Ratio Analysis

Table 14 investigates the SE bottleneck reduction ratio.

Table 14: SE reduction ratio ablation. Ratio r determines bottleneck width: FC layers have dimensions $d \to d/r \to d$.

Reduction r	**SE Params**	**F1 (%)**	**ΔF1**
2	2.1K	90.67	-0.22
4 (default)	1.1K	**90.89**	—
8	0.5K	90.52	-0.37
16	0.3K	90.21	-0.68

Reduction ratio $r = 4$ balances expressiveness with parameter efficiency. Higher ratios overly constrain the SE bottleneck, limiting channel interaction learning. Lower ratios increase parameters without improving accuracy.

F Dropout Sensitivity Analysis

Table 15 presents dropout rate sensitivity.

Table 15: Dropout rate sensitivity. Applied before classification layer.

Dropout Rate	**F1 (%)**	**ΔF1**
0.0	88.76	-2.13
0.3	90.12	-0.77
0.5 (default)	**90.89**	—
0.7	90.45	-0.44

Dropout $p = 0.5$ provides optimal regularization. Without dropout, the model overfits despite batch normalization. Higher dropout rates (0.7) excessively constrain model capacity.

G Per-Subject Detailed Results

Table 16 presents complete per-subject results for the dual-stream Kalman transformer with SE+TAP.

Table 16: Per-subject F1 scores (21-fold LOSO-CV). Subjects sorted by F1 score.

Subj	**F1**	**Prec**	**Rec**	**Subj**	**F1**	**Prec**	**Rec**
S30	98.2	98.5	97.9	S56	92.1	91.8	92.4
S44	97.8	97.2	98.4	S58	91.8	92.3	91.3
S43	97.1	96.8	97.4	S60	91.2	90.8	91.6
S45	96.5	96.9	96.1	S61	90.5	91.2	89.8
S54	95.8	95.2	96.4	S63	89.7	88.9	90.5
S55	95.3	95.7	94.9	S51	84.7	85.2	84.2
S37	94.9	94.5	95.3	S46	83.1	82.5	83.7
S38	94.2	93.8	94.6	S53	81.3	80.8	81.8
S52	93.8	94.2	93.4	S49	78.5	77.9	79.1
S36	93.1	92.7	93.5	S62	73.2	72.5	73.9
S34	92.5	92.9	92.1				
Mean				**91.10 ± 6.8**			

Performance ranges from 73.2% (S62) to 98.2% (S30). The standard deviation of 6.8% reflects inter-subject variability in fall execution style. Subjects with lower F1 scores consistently exhibit gentler fall patterns with reduced impact magnitude.

custom]

References

References

[1] World Health Organization. Falls. Fact Sheet. 2021. Available online: `https://www.who.int/news-room/fact-sheets/detail/falls` (accessed on 1 December 2024).

[2] Kalman, R.E. A New Approach to Linear Filtering and Prediction Problems. *J. Basic Eng.* **1960**, *82*, 35–45.

[3] Hu, J.; Shen, L.; Sun, G. Squeeze-and-Excitation Networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 18–22 June 2018; pp. 7132–7141.

[4] Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, L.; Polosukhin, I. Attention Is All You Need. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS), Long Beach, CA, USA, 4–9 December 2017; pp. 5998–6008.

[5] Igual, R.; Medrano, C.; Plaza, I. Challenges, Issues and Trends in Fall Detection Systems. *Biomed. Eng. Online* **2013**, *12*, 66.

[6] Ngu, A.H.; Gutierrez, Y.; Metsis, V.; Nadu, S.; Qasim, B. Personalized Watch-Based Fall Detection Using a Collaborative Edge-Cloud Framework. *Int. J. Neural Syst.* **2022**, *32*, 2250025.

[7] Madgwick, S.O.H. An Efficient Orientation Filter for Inertial and Inertial/Magnetic Sensor Arrays. Report; University of Bristol: Bristol, UK, 2010.

[8] Sabatini, A.M. Estimating Three-Dimensional Orientation of Human Body Parts by Inertial/Magnetic Sensing. *Sensors* **2011**, *11*, 1489–1525.

[9] Lin, T.Y.; Goyal, P.; Girshick, R.; He, K.; Dollár, P. Focal Loss for Dense Object Detection. In Proceedings of the IEEE International Conference on Computer Vision (ICCV), Venice, Italy, 22–29 October 2017; pp. 2980–2988.

[10] Cawley, G.C.; Talbot, N.L.C. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *J. Mach. Learn. Res.* **2010**, *11*, 2079–2107.